FastSpeech 2 Text-to-Speech for Indian English

Technical report: English model trained on the Svarah dataset

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Code and results: https://github.com/na23b005/FastSpeech2-English-Telugu

Abstract. A FastSpeech 2 model was trained from scratch on 4.76 hours of Indian English speech from the Svarah dataset (115 speakers) and validated on 0.94 hours of sentences not used in training. The ming024/FastSpeech2 implementation was used as the starting point, with a few compatibility fixes and new scripts for data preparation, training and inference. After 3 hours of training on two T4 GPUs (24897 steps) the validation loss had reached its minimum at about step 13000, and the gap between training and validation losses shows clear overfitting of the variance predictors. Of the five test sentences, E2b and E3 are transcribed correctly by an automatic speech recogniser (E3 only because the recogniser restores 'rhythm' from context), E2a has one wrong word, and E1 and E4 fail on the rare, invented or abbreviated words. The variance adaptor reproduces sentence-level prosody (declination, final falls, a pause and continuation rise at the comma) but marks lexical stress only weakly. This submission covers the English model only; the Telugu model is not included.

1. Implementation approach

1.1 Model

FastSpeech 2 [1] is a non-autoregressive acoustic model. A Transformer encoder turns the phoneme sequence into hidden vectors; the variance adaptor predicts a duration, a pitch (F0) value and an energy value for every phoneme and repeats each hidden vector for its predicted number of frames; a Transformer decoder and a PostNet then produce the 80-band mel spectrogram, which a neural vocoder converts into a waveform. In training, the true durations (from forced alignment), pitch and energy are given to the adaptor while its predictors learn to estimate them; at inference the predictions are used. Table 1 lists the components taken from the reference implementation [2]; none of the model code was changed. The model has 35.19 million parameters, not counting the vocoder.

Table 1. Components used from ming024/FastSpeech2.

Component	Configuration (English/config.yaml, as in ming024's LibriTTS recipe)	ming024 source
Encoder	4 feed-forward Transformer blocks, hidden size 256, 2 heads, convolutions of 1024 filters (kernels 9 and 1), dropout 0.2	transformer/
Speaker embedding	115 x 256, added to the encoder output	model/fastspeech2.py
Duration predictor	two Conv1D layers (256 filters, kernel 3, ReLU, layer norm, dropout 0.5) and a linear layer; predicts log(frames + 1)	model/modules.py
Pitch and energy predictors	same structure, one value per phoneme (z-score normalised); values quantised into 256 bins and embedded	model/modules.py
Length regulator	expands phoneme vectors to frame level	model/modules.py
Decoder, PostNet	6 feed-forward Transformer blocks, linear layer to 80 mel bands, 5-layer convolutional PostNet	transformer/

Component	Configuration (English/config.yaml, as in ming024's LibriTTS recipe)	ming024 source
Loss	L1 on the mel before and after the PostNet, MSE on pitch, energy and log duration	model/loss.py
Optimiser	Adam (0.9, 0.98), warm-up of 4000 steps, step annealing	model/optimizer.py
Vocoder	HiFi-GAN [4] 'universal' checkpoint, pretrained, not retrained	hifigan/

1.2 Code organisation and changes to the reference implementation

The reference code was copied into the folder FastSpeech2/ at commit d4e79eb, so that every later change can be seen in the git history. The English pipeline consists of four files, as the assignment requires: English/preprocess.py (selection, split, aligner input, features), English/train.py (training and loss plots), English/inference.py (test sentences, audio and variance-adaptor plots) and English/config.yaml. Table 2 lists all changes. Most of them are needed because the 2021 code does not run with current versions of librosa, PyTorch and the Montreal Forced Aligner.

Table 2. Changes to ming024/FastSpeech2 for the English model.

File	Change	Reason
audio/stft.py	named arguments for pad_center and librosa.filters.mel; forced .cuda() and .cpu() removed	current librosa rejects positional arguments; preprocessing without an NVIDIA GPU
preprocessor/preprocessor.py	empty TextGrid intervals read and labelled sp	MFA 3 writes silences as empty intervals; without them durations do not match the audio
preprocessor/preprocessor.py	clips without a TextGrid skipped	the original reused the previous clip's pitch and energy
utils/model.py	vocoder paths from the package folder, map_location; weights_only=False when resuming	running from the project root or on CPU; current PyTorch refuses checkpoints with numpy numbers
utils/tools.py	energy targets converted to float32	float64 energies stopped training ('Found dtype Double but expected Float')
config/LibriTTS/*.yaml	merged into English/config.yaml; Svarah paths; val_size 0; 60000 steps, annealing at 30k, 40k, 50k; checkpoint every 5000 steps; time limit of 3 h	one file per language; schedule fitted to a Kaggle session
prepare_align.py, preprocess.py	replaced by English/preprocess.py (download, filtering, selection, split, lexicon), which then calls ming024's Preprocessor	Svarah-specific preparation
train.py	English/train.py: single config, time limit, half-precision final model, --plot	Kaggle time limit; a 70 MB model fits in the repository
synthesize.py	English/inference.py: new text front end (Section 4.1), fixed sentences, one voice, plots per phoneme	ming024 used only the first dictionary pronunciation and plotted only mel spectrograms

2. Data

2.1 Dataset

Svarah [3] contains 6656 clips (9.6 h) of Indian English from 117 speakers in 65 districts, recorded on phones: about 1.4 h of read sentences, 1.7 h of read prompts and 6.4 h of spontaneous speech. It was collected for speech recognition, so recording conditions vary and each speaker contributes only about five minutes. One multi-speaker model with a speaker embedding was therefore trained.

2.2 Selection and subsampling

Svarah has no speaker identifier, so a speaker was defined as a unique combination of the speaker metadata (gender, age group, first language, state, district, education, job category and occupation domain). This gives 115 speakers in the selected data. People with identical metadata are merged by this rule; the largest such speaker (spk073, 14.5 min) is probably several people.

Clips were kept if they last 1 to 11 s (the model handles at most 1000 frames, 11.6 s) and if the transcript contains only letters and basic punctuation; this removes transcripts with digits, symbols and annotation tags, which neither the lexicon nor the aligner can map to phonemes reliably. The remaining clips were shuffled with a fixed seed and taken until 6 h were reached. Only 4212 clips (5.70 h) passed the filter, so all of them were used.

2.3 Train and validation split

A fixed hash of each normalised sentence assigns one in six distinct sentences to validation. All recordings of a sentence fall on the same side, so no clip and no sentence occurs in both sets. The script also checks that every validation speaker occurs in training, since a multi-speaker model can only produce voices it has learned. The full list of clips is stored in English/data/split.csv.

Table 3. Final split.

Set	Clips	Hours	Speakers	Training audio per speaker
Training	3532	4.76	115	median 1.7 min, maximum 14.5 min
Validation	680	0.94	107, all present in training	

2.4 Preprocessing

Each clip was converted to mono, resampled to 22050 Hz, peak-normalised and stored as a 16-bit wav file together with its transcript, cleaned with ming024's english_cleaners. The audio settings follow the pretrained vocoder: FFT size 1024, hop 256 samples (11.6 ms per frame), 80 mel bands between 0 and 8000 Hz. The pronunciation dictionary is ming024's LibriSpeech lexicon (ARPAbet with stress marks), extended with g2p_en pronunciations for words it does not contain, mostly Indian names and places.

Phoneme durations were obtained with the Montreal Forced Aligner 3 [5] and its english_us_arpa acoustic model. The beam was widened to 100 (retry beam 400) because phone recordings and Indian accents fit the US model less well; the second pass used speaker adaptation (fMLLR). Six dictionary pronunciations containing a phone unknown to the acoustic model were ignored. ming024's Preprocessor then trimmed leading and trailing silence and computed, for every phoneme, its duration in frames, its mean pitch (pyworld, linearly interpolated through unvoiced frames) and its mean energy (L2 norm of the STFT frame). Pitch and energy were z-score normalised; the statistics are stored in stats.json (pitch mean 177.8 Hz, standard deviation 56.7 Hz).

3. Training

3.1 Configuration

Table 4. Training configuration.

Setting	Value
Hardware	Kaggle notebook with two NVIDIA T4 GPUs (nn.DataParallel), Python 3.13
Batch	16 clips; groups of four batches are sorted by length to reduce padding
Learning rate	256^-0.5 x min(step^-0.5, step x 4000^-1.5): peak 9.9e-4 at step 4000, 4.0e-4 at the end; multiplied by 0.3 after steps 30000, 40000 and 50000 (not reached)

Setting	Value
Regularisation	dropout 0.2 in the Transformer and 0.5 in the variance predictors; gradient norm clipped at 1.0
Length	at most 60000 steps or 3 h; stopped by the time limit at step 24897 (about 2.3 steps per second)
Monitoring	training loss every 100 steps; validation loss on all 680 clips every 1000 steps
Final model	weights at the last step, saved in half precision (70 MB)

Training was run three times with the same settings. The first run was lost with its Kaggle session, and in the second Kaggle job the training was accidentally executed twice in a row. The reported model and curves come from the last training. The three runs ended at almost the same validation loss (2.536 at step 26000, 2.543 at step 25000 and 2.536 at step 24000), so the result is repeatable.

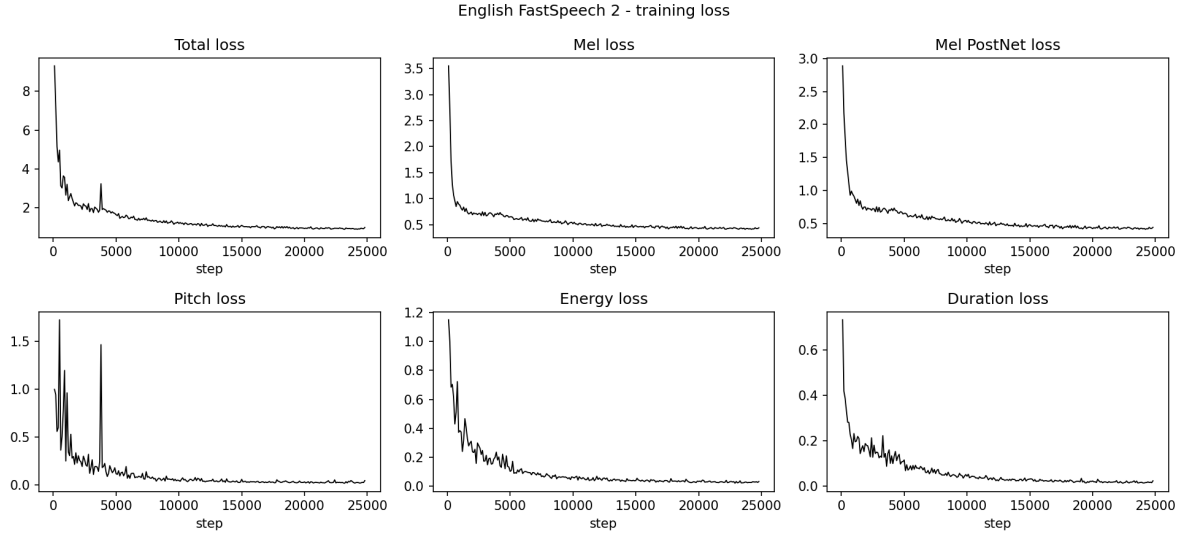


Figure 1. Training losses, logged every 100 steps.

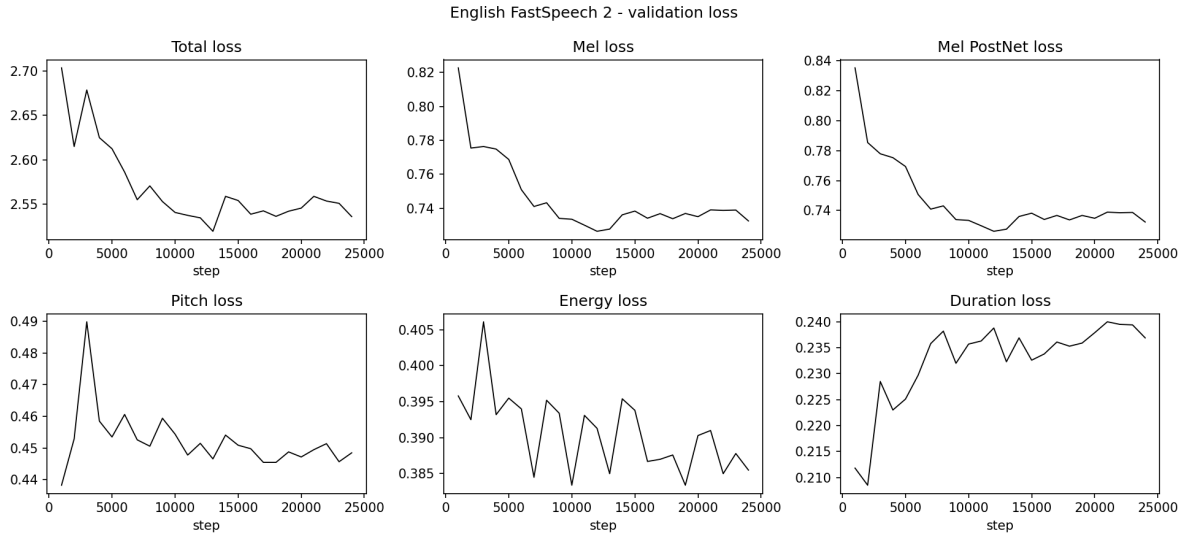


Figure 2. Validation losses on the 680 validation clips, every 1000 steps (y-axis ranges differ).

Table 5. Loss values at selected steps.

Loss	Train, step 100	Train, step 24800	Val., step 1000	Val., best (step)	Val., step 24000
Total	9.316	0.979	2.704	2.519 (13000)	2.536
Mel	3.550	0.441	0.823	0.727 (12000)	0.733
Mel after PostNet	2.887	0.441	0.835	0.726 (12000)	0.732
Pitch	0.996	0.043	0.438	0.438 (1000)	0.449
Energy	1.151	0.031	0.396	0.383 (10000)	0.386
Duration	0.733	0.022	0.212	0.209 (2000)	0.237

3.2 Analysis of the loss curves

The training losses fall steeply during the first 1000 steps and slowly afterwards; at step 24800 they were still decreasing, so training had not fully converged. The mel losses make up most of the total. Two short spikes occur during warm-up, both caused by the pitch loss: at step 500 (pitch 1.72, total 4.97) and at step 3800 (pitch 1.46, total 3.24), as the learning rate approached its peak at step 4000. Both disappeared within 100 steps, and the curves are smooth after warm-up.

The validation curves show overfitting. The validation mel loss improves until about step 12000 and stays flat afterwards, and the total validation loss is lowest at step 13000. The validation pitch and energy losses do not improve after the first evaluation, and the validation duration loss rises from 0.21 to 0.24. At the end the training pitch, energy and duration losses are 10 to 20 times lower than the validation values, which means the variance predictors have memorised the training sentences rather than learned rules that carry over to new text. This is not surprising for 4.76 h of mostly spontaneous speech with a median of 1.7 min per speaker. Because of the time limit the learning rate was never annealed (the first step was set at 30000). A checkpoint around step 13000 would give practically the same validation loss (2.52 against 2.54).

4. Generated speech

4.1 Text processing and voice

The test sentences were used exactly as given. English/inference.py converts them to phonemes with g2p_en, which looks words up in the CMU dictionary, chooses the pronunciation of homographs from the part-of-speech tag and predicts unknown words with a neural model. Four corrections were added after inspecting the output:

(1) '10:45' is split into '10 45'; otherwise the colon is dropped and the time is read as one word.

(2) numbers are written out with ming024's normalize_numbers ('12' as 'twelve', '2026' as 'twenty twenty-six') and hyphens are replaced by spaces, because g2p_en mispronounces hyphenated words such as 'forty-five';

(3) 'AM' is spelled as two letters (EY1 EH1 M) instead of being read as the word 'am';

(4) a short pause is inserted before each sentence. Without it the first 'The' received only 6 frames (70 ms) and was not audible in E2a for any of the voices tried; with it the first word gets 9 to 17 frames in all sentences.

The voice is training speaker spk029 (female, 9.6 min of training audio), chosen by listening to four speakers and an earlier checkpoint. It sounded better than spk073, the speaker with the most data.

4.2 Observations for each test sentence

The observations combine listening with an automatic check: each file was transcribed with the Whisper small.en recogniser [7] (faster-whisper implementation, CPU). Whisper's language model corrects near-misses (it writes 'rhythm' for E3), so a correct transcript shows that the words are recognisable in context, not that every sound is right. When listening, the speech was judged partially correct overall; the clearest problem, the missing first 'The', was fixed as described above.

Table 6. Generated speech for the five test sentences.

ID	Phonemes of interest	Whisper transcript (errors underlined)	Remarks
E1	antidisestablishmentarianism: AE2 N T AY0 D IH0 S AH0 S T AE2 B L IH0 SH M AH0 N T EH1 R IY0 AH0 N IH0 Z AH0 M (CMU dictionary)	The high tide is establishmentarianism movement, the inconsiderable momentum during the parliamentary proceedings.	The second half of the long word is recognised, its beginning is not; 'gained' is lost. The word is spoken at an even, fast pace (1.64 s for 28 phonemes).
E2a	conduct (noun): K AA1 N D AH0 K T	The orchestra's conduct during the ceremony was pressed by the audience.	Correct noun stress in the input. 'praised' is heard as 'pressed'.
E2b	conduct (verb): K AA0 N D AH1 K T	The researchers will conduct an experiment to evaluate the proposed method.	Fully recognised. Correct verb stress in the input, so the two readings of 'conduct' differ.
E3	rhyth: R AY1 DH (predicted); m: EH1 M (letter name)	The rhythm of the orchestra changed unexpectedly.	Since the text is kept as given, 'rhyth m' is read as 'rithe' followed by the letter M; Whisper restores 'rhythm' from context. The rest is intelligible.
E4	ten forty five; AM: EY1 EH1 M; twelve November twenty twenty six; andaroo: AE0 N D AA1 R OW0 (predicted)	The meeting will start at 10.45 pm on 12th November 2026, so in the rush to arrive on them.	Time and date are recognised. 'AM' is heard as 'pm'; 'andaroo should' and 'time' are not recognised. The date is read 'twelve November', as written.

Overall, common words are recognisable, and the errors are concentrated on rare, long or invented words and on short words and letters (the first 'The', 'AM') that receive too few frames. No formal listening test (for example a mean opinion score) was carried out.

5. Variance adaptor

Figures 3 to 7 show the predicted duration (in frames of 11.6 ms), pitch (in Hz, de-normalised with stats.json) and energy of every input phoneme; dotted lines mark word boundaries. The separate plots required by the assignment are in Results/English/variance_adaptor_plots/. Pitch values for pauses and voiceless consonants (P, T, K, S and others) are not heard: the model was trained on F0 interpolated through unvoiced frames, so these values only continue the contour.

5.1 General patterns

Table 7. Summary of the predicted values.

Sentence	Length (s)	Mean frames, vowels / consonants	Mean pitch, first / last third (Hz)	Pitch of last phoneme (Hz)	Mean energy, vowels / consonants
E1	6.03	4.9 / 5.9	213 / 198	230 (rise)	85 / 48
E2a	4.11	5.8 / 6.7	216 / 180	130 (fall)	91 / 41
E2b	4.20	5.5 / 6.0	216 / 190	174 (fall)	96 / 40
E3	3.16	7.0 / 6.3	227 / 191	185 (fall)	84 / 37
E4	6.88	5.9 / 7.1	221 / 190	139 (fall)	86 / 42

Duration. Most phonemes receive 3 to 9 frames. Sibilants and affricates (S, Z, SH, CH) are among the longest (10 to 13 frames), and the last phoneme of each sentence is lengthened (Z 11 frames in

E1, S 13 in E2a, IY0 17 in E3). Stressed and unstressed vowels have nearly the same average duration (5.1 against 4.8 frames in E1), so the duration predictor reflects phoneme identity and position in the phrase but hardly lexical stress.

Pitch. Every sentence starts high and falls gradually (first third 213 to 227 Hz, last third 180 to 198 Hz), the usual declination of declarative sentences. E2a, E2b, E3 and E4 end with a fall; E1 ends with a rise on 'proceedings'. The mean of 199 to 207 Hz fits the female voice (the mean over all training speakers is 178 Hz). Stressed vowels are not consistently higher than unstressed ones.

Energy. Vowels have about twice the energy of consonants, voiceless stops and fricatives are the weakest (about 4 to 25) and pauses are close to zero. The energy predictor therefore follows the acoustic class of each phoneme closely and is the most regular of the three predictors.

5.2 E1: a long, rare word

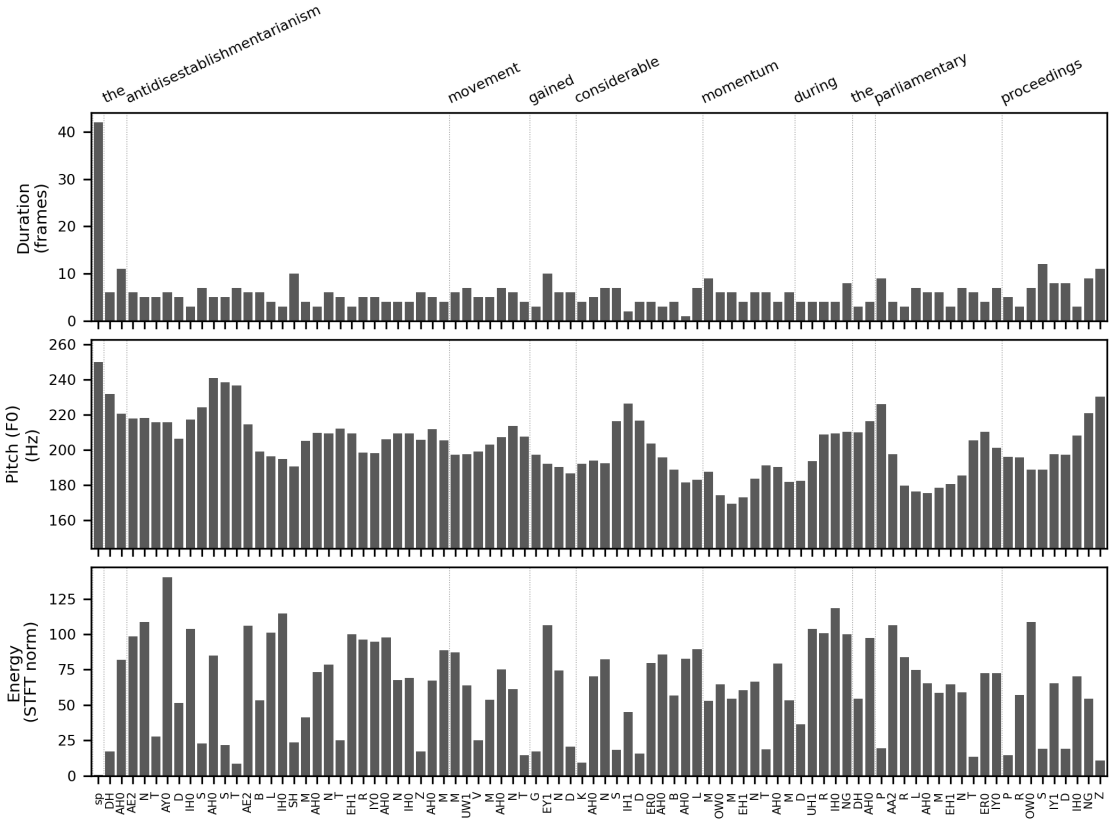


Figure 3. E1: predicted duration, pitch and energy per phoneme (dotted lines: word boundaries).

The word antidisestablishmentarianism takes 141 frames (1.64 s) for 28 phonemes, an almost uniform 3 to 7 frames each. The vowel with primary stress (EH1 in '-tar-') gets only 3 frames and no pitch peak (209 Hz). The highest pitch (238 to 241 Hz) lies on the unstressed 'dis-es' near the start of the word. The contour then falls through 'movement', 'considerable' and 'momentum' to about 170 Hz and rises again on 'parliamentary' and at the end of 'proceedings' (221 to 230 Hz). The even timing and the misplaced peak are consistent with the recogniser missing the beginning of the word. 'gained' has a long vowel (EY1, 10 frames) but very weak consonants (energy 17 for G and 20 for D).

5.3 E2a: 'conduct' as a noun

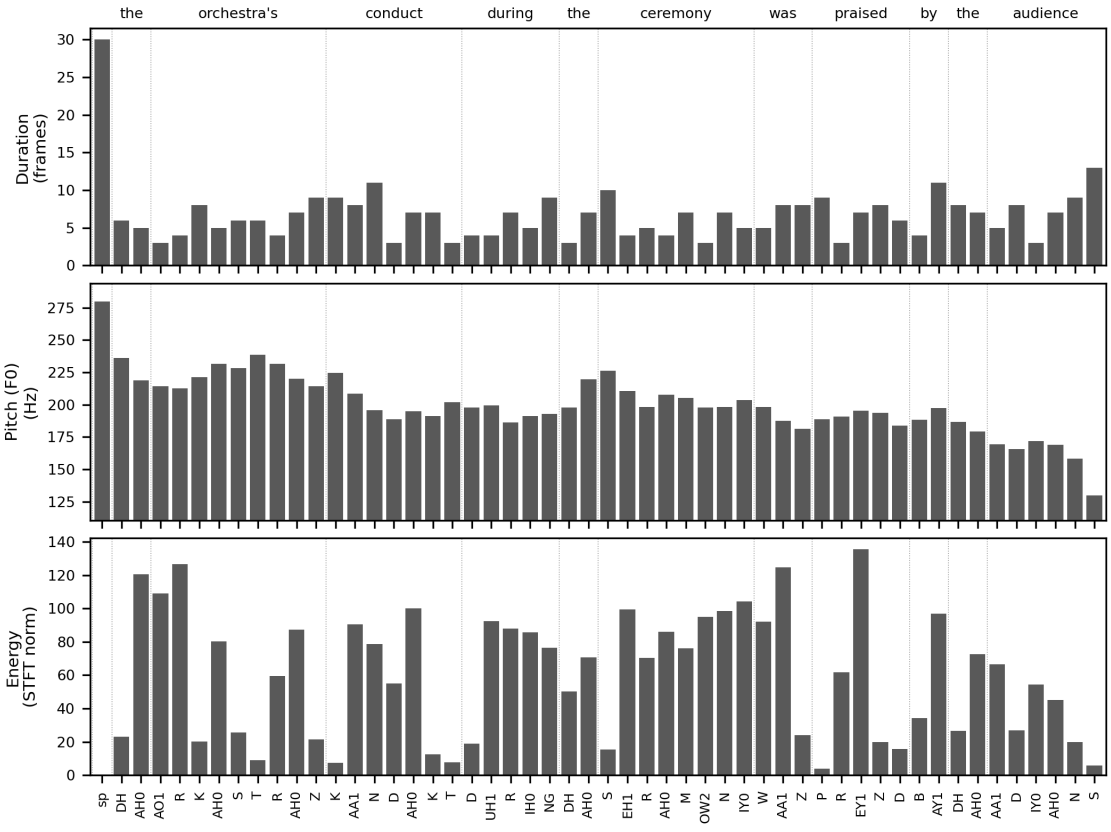


Figure 4. E2a: predicted duration, pitch and energy per phoneme (dotted lines: word boundaries).

In the noun the first syllable is slightly longer (AA1 8 frames against AH0 7) and higher (208 against 195 Hz), in line with initial stress, although the second vowel is louder (100 against 91). This sentence has the clearest declination: about 230 Hz on 'orchestra's', 200 Hz in the middle, then 158 Hz and finally 130 Hz on the closing N and S of 'audience'. The final S is also the longest phoneme (13 frames). 'praised' has the highest energy in the sentence (EY1, 135), but its diphthong lasts only 7 frames, which may be why it is heard as 'pressed'.

5.4 E2b: 'conduct' as a verb

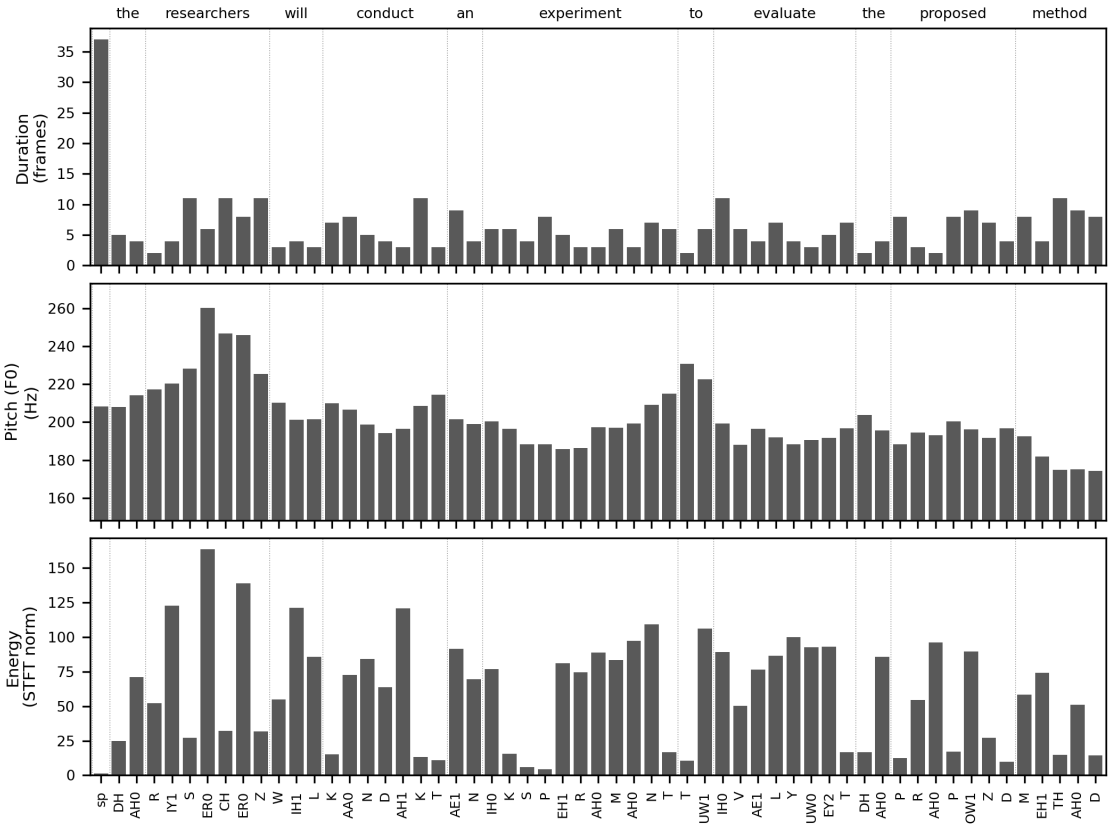


Figure 5. E2b: predicted duration, pitch and energy per phoneme (dotted lines: word boundaries).

In the verb the stressed second vowel AH1 is much louder than the first (121 against 73) but shorter (3 against 8 frames) and not higher (197 against 206 Hz). Together with E2a this shows that the model separates the two readings mainly by vowel quality (AA against AH) and energy, not by duration or pitch. The pitch maximum (260 Hz) falls on 'researchers', the first content word, and the contour declines to 174 Hz on 'method', whose last three phonemes are lengthened (TH 11, AH0 9, D 8 frames).

5.5 E3: the split word 'rhyth m'

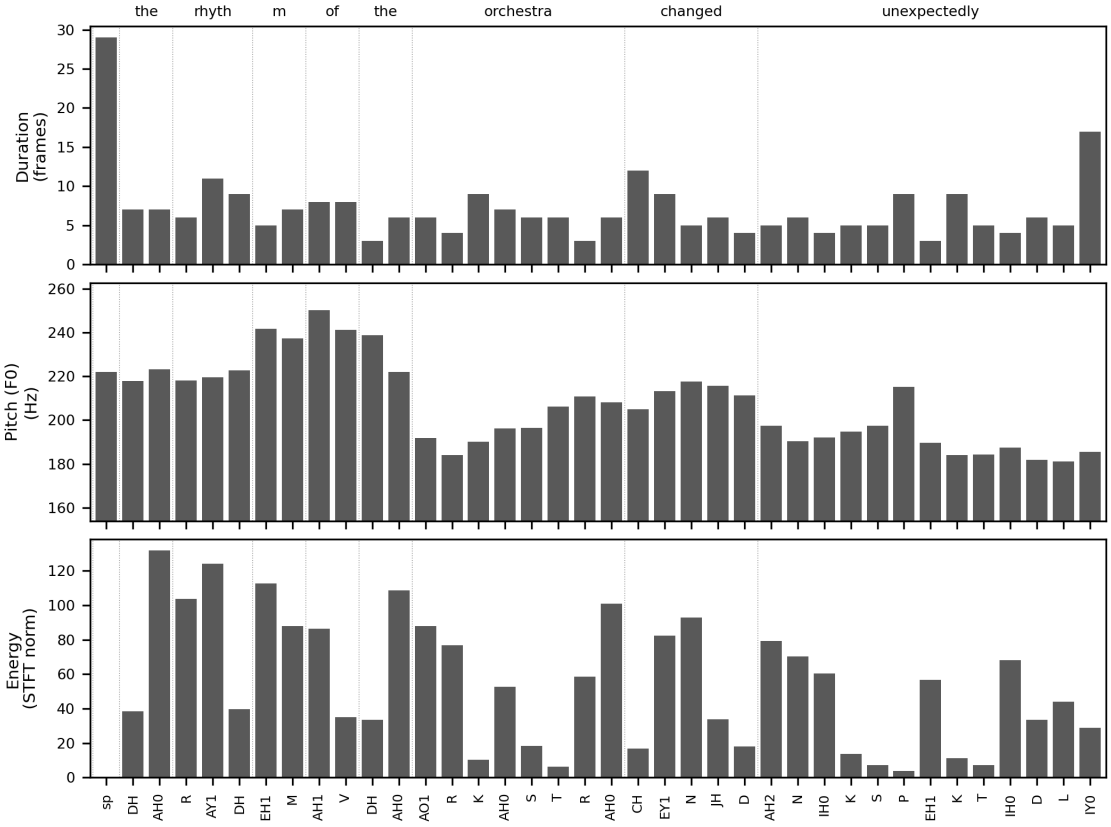


Figure 6. E3: predicted duration, pitch and energy per phoneme (dotted lines: word boundaries).

'rhyth' is read with a long diphthong (AY1, 11 frames), and the separate letter M (EH1 M) carries the pitch maximum of the sentence (242 to 250 Hz on 'm of'), so the model treats the stray letter as a prominent word. From 'the orchestra' onward the pitch stays between about 180 and 220 Hz. The final IY0 of 'unexpectedly' shows the strongest final lengthening of all sentences (17 frames, 0.2 s) with low energy (29).

5.6 E4: time, date and an invented word

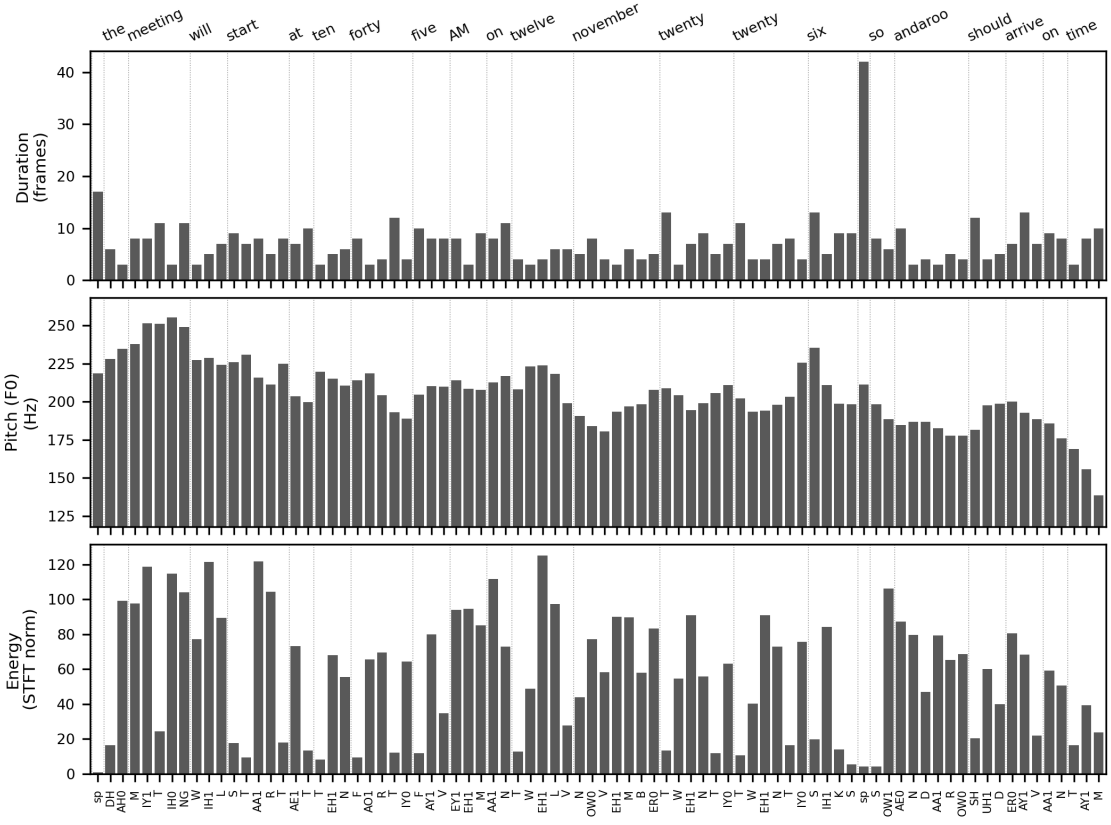


Figure 7. E4: predicted duration, pitch and energy per phoneme (dotted lines: word boundaries).

The comma after 'twenty six' becomes a pause of 42 frames (0.49 s), preceded by a continuation rise (226 Hz on IY0, 235 Hz on the S of 'six'). After the pause the pitch falls steadily to 139 Hz on the final M of 'time', the lowest value in any sentence. In 'AM' (EY1 8, EH1 3, M 9 frames) the second letter is very short, which agrees with it being heard as 'pm'. In 'andaroo' the unstressed first vowel AE0 is the longest phoneme (10 frames) while the stressed AA1 gets only 3, so the stress of the predicted pronunciation is not realised.

6. Limitations and possible improvements

No comparison with Telugu can be made because the Telugu model is not part of this submission. Table 8 summarises the failures seen in the English test sentences, their likely causes and possible remedies.

Table 8. Failures, causes and improvements.

Observation	Likely cause	Possible improvement
Beginning of 'antidisestablishmentarianism', 'andaroo' and 'AM' not recognised; even timing in long words	little data (4.76 h); weak modelling of lexical stress; no training examples of rare or invented words	pre-training on a larger English TTS corpus (LJSpeech, LibriTTS) followed by fine-tuning on Svarah
Short words and letters too short (first 'The', 'AM')	reduced function words in spontaneous speech; clip starts trimmed	leading pause (applied); read-speech subset; word-level duration control
Validation losses of the variance predictors do not improve	median 1.7 min per speaker, varied phone recordings	early stopping, stronger regularisation, data augmentation, cleaner read-speech subset
Speech judged only partially correct when listening	small and varied training data; vocoder never trained on Svarah audio	more and cleaner data; fine-tuning HiFi-GAN on Svarah mel spectrograms

Observation	Likely cause	Possible improvement
Some voices less clear (spk073)	speakers derived from metadata merge different people	speaker clustering with speaker embeddings
Learning rate never annealed	time limit reached at step 24897	annealing steps within reach, e.g. 12500, 17500 and 22500
'12 November' read as 'twelve November'	only cardinal numbers handled	date-aware text normalisation

7. Reproducibility

requirements.txt lists the Python packages with the versions used for inference; the Montreal Forced Aligner is installed from conda-forge with micromamba. Preprocessing and training were run on Kaggle (Python 3.13, two T4 GPUs), inference on a laptop CPU (Python 3.12, PyTorch 2.14.1). All settings are in English/config.yaml. Data selection and split are deterministic (fixed seed and hash); training is not seeded, but two runs gave validation losses within 0.01. Svarah is a gated dataset, and the Hugging Face token is read from the HF_TOKEN environment variable. The commands, run from the repository root (on Kaggle prefixed with '!'), are:

```
python English/preprocess.py prepare
micromamba create -p /tmp/mfa -c conda-forge montreal-forced-aligner
mfa model download acoustic english_us_arpa
mfa align English/data/raw English/data/lexicon.txt english_us_arpa \
    English/data/preprocessed/TextGrid -j 4 --clean --beam 100 --retry_beam 400
python English/preprocess.py features
unzip FastSpeech2/hifigan/generator_universal.pth.tar.zip -d FastSpeech2/hifigan
python English/train.py   (--restore_step N resumes from a checkpoint)
python English/train.py --plot
python English/inference.py
```

On Kaggle, preparation took about 10 min, the aligner installation 5 min, alignment about 30 min, feature extraction about 20 min and training 3 h. Inference takes a few seconds per sentence on a CPU. The exact notebook cells are given in README.md.

References

[1] Y. Ren, C. Hu, X. Tan, T. Qin, S. Zhao, Z. Zhao and T.-Y. Liu, "FastSpeech 2: Fast and high-quality end-to-end text to speech," in Proc. ICLR, 2021.

[2] C.-M. Chien (ming024), FastSpeech2, https://github.com/ming024/FastSpeech2, commit d4e79eb.

[3] T. Javed et al., "Svarah: Evaluating English ASR systems on Indian accents," in Proc. Interspeech, 2023; data: https://huggingface.co/datasets/ai4bharat/Svarah.

[4] J. Kong, J. Kim and J. Bae, "HiFi-GAN: Generative adversarial networks for efficient and high fidelity speech synthesis," in Proc. NeurIPS, 2020.

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[7] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in Proc. ICML, 2023.